IGCSE 0607 Practice Questions — E2.1

Introduction to algebra

Total Questions: 30 — **Total Marks:** 131 Marks

Table of Contents & Quick Navigation

- **E2.1** Introduction to algebra
  → **Paper 2 (Non-Calculator)** *(18 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

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1 Work out the value of the expression

$3x^2 - 5xy$

when $x = -4$ and $y = 2$.

..... **[2]**

2 Work out the value of

$5x^2 - 3y$

when $x = -2$ and $y = 4$.

..... **[2]**

3 Find the value of $3a - 2b$ when $a = 4$ and $b = -5$.

..... **[2]**

4 Work out the value of

$$3x^2 - 4y$$

when $x = -3$ and $y = -5$.

..... **[2]**

5 Work out the value of

$$\frac{3x - y^2}{2z}$$

when $x = 5$, $y = -4$ and $z = 0.5$.

..... [**2**]

6 Given the formula $v = u + at$.

Find the value of v when $u = 5$, $a = -2$, and $t = 3$.

$v =$ ..... **[3]**

7 **(a)** Work out the value of $4x - 3y$ when $x = -2$ and $y = 5$.

..... **[2]**

(b) Work out the value of $\dfrac{p^2 - 7}{2q}$ when $p = 5$ and $q = -3$.

..... **[2]**

8 **(a)** Work out the value of $3a - 5b$ when $a = -2$ and $b = 4$.

..... **[2]**

(b) Work out the value of $x^2 + 4x$ when $x = -3$.

..... **[2]**

9 Consider the formula

$$P = 3a^2 - 2ab + 5b$$

(a) Calculate the value of P when $a = 4$ and $b = -3$.

$P =$ ..... **[2]**

(b) Calculate the value of b when $P = 19$ and $a = -2$.

$b =$ ..... **[2]**

10 **(a)** Work out the value of $5x - 2y$ when $x = 3$ and $y = -4$.

..... **[2]**

(b) Evaluate

$$\frac{a^2 - b^3}{2c}$$

when $a = -4$, $b = 2$ and $c = 3$.

..... **[2]**

11 **(a)** **(a)** The cost of hiring a plumber is calculated using the formula

$C = 40 + 25h$ where C is the cost in dollars and h is the number of hours worked.
Work out the cost of hiring a plumber for 3.5 hours.

$C = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

(b) Given that $v = u + at$, find the value of v when $u = -12$, $a = 9.8$ and $t = 3$.

$v = \ldots\ldots\ldots\ldots\ldots$ **[3]**

12 **(a)** A plumber charges a fixed fee of \$40 plus \$25 for each hour he works.
Write down an expression, in terms of h, for the total cost, in dollars, for h hours of work.

\$ ..... **[1]**

(b) Given the formula

$$P = \frac{k^3 - m^2}{\sqrt{n}}$$

Work out the exact value of P when $k = -2$, $m = -3$ and $n = \frac{1}{16}$.

$P =$ ..... **[3]**

13 Find the exact value of

$$W = \frac{p^{-2} + q^3}{\sqrt[3]{r}}$$

when $p = \frac{1}{2}$, $q = -2$ and $r = -\frac{1}{8}$.

$W =$ ..... **[4]**

14 The distance, d, between two points (x_1, y_1) and (x_2, y_2) is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(a) Work out the value of d when $x_1 = 4$, $y_1 = -3$, $x_2 = -8$, and $y_2 = 2$.

$d =$ ..... **[2]**

(b) Given that $d = 17$, $x_1 = k$, $y_1 = 8$, $x_2 = 3$, and $y_2 = -7$, work out the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

15 The nth term of a sequence is given by the expression

$$T_n = an^2 + bn + c$$

where a, b, and c are constants.
The first three terms of the sequence are $T_1 = 4$, $T_2 = 13$, and $T_3 = 28$.

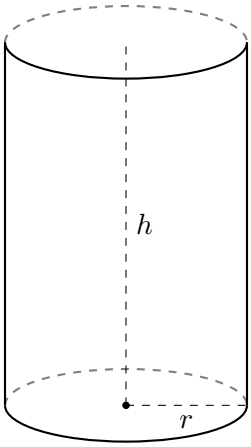
(a) By substituting $n = 1$, $n = 2$, and $n = 3$ into the expression, find the values of a, b, and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

(b) Find the value of k such that $T_{k+1} - T_k = 57$.

$k =$ ..... **[2]**

16 A cylindrical container of radius r and height h is open at the top.
The total exterior surface area of the container, A, consists of the curved surface area and the circular base.



(a) Show that the total surface area A can be written as

$$A = \pi r(2h + r)$$

[2]

(b) **(i)** Calculate the exact surface area A when $r = 4$ and $h = 6$.
Give your answer in terms of π.

$A =$ ..... **[2]**

(ii) The volume of the cylinder is V.
The formula for the volume is $V = \pi r^2 h$.
Express h in terms of V, π and r, and substitute this into the expression for A to show that

$$A = \frac{2V}{r} + \pi r^2$$

[3]

17 Consider the expression $P = \dfrac{2x^2 - 3y}{x - y}$.

(a) Calculate the value of P when $x = -2$ and $y = 4$.

$P =$ ..... **[3]**

(b) Rearrange the formula to make y the subject.

$y =$ ..... **[4]**

18 The mathematical operation $*$ is defined for all non-zero numbers a and b as

$$a * b = \frac{a^2 - b^2}{ab}$$

(a) Evaluate the following, giving each answer as a fraction in its simplest form.

(i) $6 * (-2)$

..... **[2]**

(ii) $\frac{1}{2} * 4$

..... **[3]**

(b) Given that $x * 2 = \frac{3}{2}$, find the positive value of x.

$x =$ ..... **[3]**

19 A plumber charges a fixed call-out fee of $\$C$ and an hourly rate of $\$R$.

(a) Write down an expression for the total cost, T dollars, for a job that takes h hours.

$T =$ ..... **[1]**

(b) The call-out fee is \$45 and the hourly rate is \$32.50.
Work out the total cost for a job that takes 3.5 hours.

\$ ..... **[2]**

20 A formula is given by

$$P = \frac{\sqrt{x - y}}{z^3}$$

Calculate the value of P when $x = 12.5$, $y = 3.84$, and $z = -1.2$.
Give your answer correct to 3 significant figures.

$P =$ ..... **[3]**

21 A solid shape has a volume, V, given by the formula

$$V = \pi r^2 h + \frac{2}{3}\pi r^3$$

Work out the value of V when $r = 4.5$ and $h = 8.2$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

22 **(a)** A plumber charges a fixed fee of \$45 plus \$32 for each hour worked.
Write down a formula for the total charge, C dollars, when the plumber works for h hours.

$C =$ ..... **[1]**

(b) Work out the total charge when the plumber works for 3.5 hours.

\$ ..... **[1]**

(c) Work out the value of

$$x^3 - 4xy$$

when $x = -3$ and $y = 2.5$.

..... **[2]**

23 The volume, V, of a cone with radius r and height h is given by the formula

$$V = \frac{1}{3}\pi r^2 h$$

(a) Calculate the value of V when $r = 6.4$ and $h = 12.6$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[2]**

(b) Work out the value of r when $V = 800$ and $h = 14$.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

24 The cost, C dollars, of hiring a campervan for d days and travelling m kilometres is given by the formula

$$C = 85d + 0.24m$$

(a) Calculate the cost of hiring the campervan for 7 days and travelling 450 kilometres.

\$ ..... **[2]**

(b) A family hires the campervan for 14 days.
The total cost is \$1475.60.
Work out the number of kilometres they travelled.

..... km **[3]**

25 The cost, C dollars, of hiring a plumber for h hours is given by the formula:

$C = 45 + 35h$

(a) Calculate the cost of hiring the plumber for 3.5 hours.

$C =$ ..... **[2]**

(b) The plumber charges a total of \$325 for a job.
Calculate the number of hours the plumber worked.

$h =$ ..... hours **[3]**

26 The total surface area, A cm^2, of a solid cone with base radius r cm and vertical height h cm is given by the formula

$$A = \pi r \left(r + \sqrt{r^2 + h^2} \right)$$

(a) Calculate the value of A when $r = 5$ and $h = 12$.
Give your answer correct to 3 significant figures.

$A =$ ..... **[2]**

(b) Another solid cone has base radius $r = 6$ and total surface area $A = 96\pi$.
Calculate the vertical height, h, of this cone.

$h =$ ..... **[4]**

27 The volume, V, of a spherical cap of height h and base radius a is given by the formula

$$V = \frac{1}{6}\pi h(3a^2 + h^2)$$

(a) Calculate the volume of a spherical cap with height $h = 4.2$ cm and base radius $a = 6.5$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

(b) Rearrange the formula to make a^2 the subject.

$a^2 =$ ..... **[3]**

(c) Find the base radius, a, of a spherical cap with volume $V = 150$ cm^3 and height $h = 3$ cm. Give your answer correct to 3 significant figures.

$a =$ ..... cm **[1]**

28 The value of y is given by the formula

$$y = \frac{a - b^2}{c - \sqrt{d}}$$

Work out the value of y when $a = -4.5$, $b = -3.2$, $c = 1.8$ and $d = 0.64$.

$y =$ ..... **[3]**

29 A formula used in physics is

$$s = ut + \frac{1}{2}at^2$$

(a) **(a)** Calculate the value of s when $u = 5.2$, $t = 4.5$ and $a = -9.8$.

$s =$ ..... **[2]**

(b) Calculate the positive value of t when $s = -9.8$, $u = 4.9$ and $a = -9.8$.

$t =$ ..... **[3]**

30 The volume, V, and surface area, A, of a solid are given by the formulas below.

(a) The formula for the volume is

$$V = \frac{1}{3}\pi h(r^2 + rR + R^2)$$

Calculate the value of V when $h = 12.5$, $r = 4.2$, and $R = 7.8$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

(b) The formula for the surface area is

$$A = \pi(R + r)\sqrt{(R - r)^2 + h^2} + \pi r^2 + \pi R^2$$

Calculate the value of A when $h = 12.5$, $r = 4.2$, and $R = 7.8$.
Give your answer correct to 1 decimal place.

$A =$ ..... **[3]**